

Hamiltonian Cycles in Double Path Graphs

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Abstract

This thesis seeks to increase our understanding of the Hamiltonian problem by restricting its input to double path graphs, or graphs consisting of two edge-disjoint spanning paths.

1 Introduction

The Hamiltonian problem is a very old, well-known problem in graph theory. Although the problem is NP-complete, many researchers have tackled it since its conception, narrowing the difficulty of the problem by finding conditions for which graphs are always Hamiltonian or never Hamiltonian. The history of the problem is quite vast, and although this introduction will cover some important results, a full survey of the Hamiltonian problem is far beyond the scope of this thesis.

Among the many ways researchers have expanded the known conditions of the Hamiltonian problem is by restricting the input to certain types of graphs. This is the approach this thesis takes, by restricting input to the Hamiltonian problem to double path graphs.

Before we enter into a discussion of Hamiltonian cycles and double path graphs, it is necessary to define a few terms and basic concepts. This introduction will first discuss some common terminology in graph theory, then define the Hamiltonian problem and provide some related history.

1.1 Terminology

A graph G consists of a finite nonempty set of **vertices**, $V(G)$, and a finite set of **edges**, $E(G)$. Each edge in $E(G)$ **joins** a pair of vertices in $V(G)$. An

edge between vertices a and b is represented as ab . The graph shown below in Figure 1.1 has 5 vertices, a , b , c , d , and e , and 5 edges, ab , bc , bd , cd , and de .

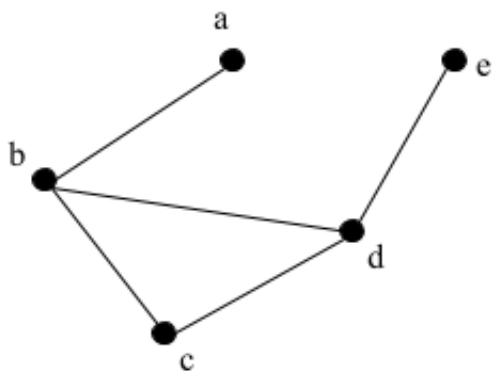


Figure 1.1: A graph to demonstrate basic terminology.

Two vertices that are joined by an edge are said to be **adjacent**. In Figure 1.1, a and b are adjacent, while c and e are nonadjacent. A vertex is **incident** to an edge if the vertex is one of the endpoints of that edge. An edge is also considered incident to a vertex if the vertex is one of its endpoints. In Figure 1.1, vertex b is incident to edge bc , and bc is incident to b .

The **order** of a graph is the number of vertices it contains, $|V(G)|$, often represented as n . The **size** of a graph is the number of edges it contains, $|E(G)|$, often represented as q . The graph shown below in Figure 1.2 has order $n = 5$ and size $q = 7$.

The **degree** of a vertex v , $\deg v$, is the number of edges to which v is incident. This is equivalent to the number of vertices to which it is adjacent. The **minimum degree** of G , $\delta(G)$, is the degree of the lowest-degree vertex in G . The **maximum degree** of G , $\Delta(G)$, is the degree of the highest-degree vertex in G .

Two vertices a and b in a graph G are **connected** if there is a path from a to b in G . A graph G is also **connected** if every pair of vertices in G are connected. A graph that is not connected is said to be **disconnected**. If G is disconnected, each connected part of G is a **component** of G .

If there is a vertex v in a connected graph G which, if removed, would cause G to be disconnected, then v is a **cut vertex**. If there is a set of

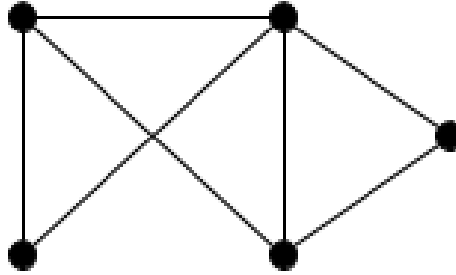


Figure 1.2: A graph to demonstrate order and size.

vertices V_1 in a connected graph G which, if removed, would cause G to be disconnected, then V_1 is a **cut set**.

A **walk** is a sequence of vertices $v_1, v_2, \dots, v_k$ such that v_{i+1} is adjacent to v_i for any $1 \leq i < k$. A walk in which no edge is repeated is a **trail**, and a walk in which no vertex is repeated is called a **path**. No walk can repeat an edge without also repeating a vertex, so we can see that every path is also a trail, but not every trail is a path. In Figure 1.3, the walk a, c, d, b, c, e is a trail, as it repeats no edges, but not a walk, as the vertex c is repeated.

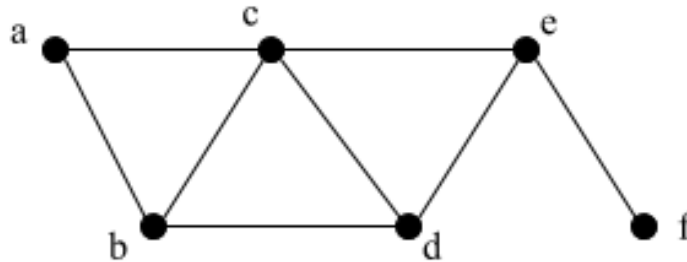


Figure 1.3: A graph to demonstrate walks, trails, paths, circuits, and cycles.

A **circuit** is a walk which contains at least three edges, begins and ends on the same vertex, and does not repeat any edges. A **cycle** is a walk which contains at least three vertices, begins and ends on the same vertex, and does not repeat any vertices, aside from the last vertex. Similarly, we can see that every cycle is a circuit, but not every circuit is a cycle. In Figure 1.3, the walk a, c, e, d, c, b, a is a circuit, but not a cycle, as c is once again repeated. The walk a, b, d, e, c, a is both a circuit and a cycle. Although the vertex a

is repeated, this is permitted in a cycle because it is the first vertex, and is only repeated as the last vertex.

A **subgraph** of a graph G is a graph which is composed of a subset of the vertices $V(G)$ and a subset of the edges $E(G)$. Given a subset E_1 of the edges in G , we denote $\langle E_1 \rangle$ as the subgraph induced by E_1 .

1.2 The Hamiltonian Problem

A **Hamiltonian path** in a graph G is a path that contains every vertex in G , or a spanning path. A **Hamiltonian cycle** in a graph G is a cycle that contains every vertex in G , or a spanning cycle. A graph is said to be **Hamiltonian** if it contains a Hamiltonian cycle, and **semi-Hamiltonian**, or **traceable**, if it contains a Hamiltonian path.

1.2.1 History of the Hamiltonian Problem

The Hamiltonian problem is named for William Rowan Hamilton, who invented the icosian game in the 1850s. Since then, a vast amount of research has been done into spanning paths and cycles. Gould identifies four fundamental results of this research that make up the foundation of modern work on the problem. These four fundamental results are discussed below.

Dirac's Theorem: If G is a graph of order n such that $\delta(G) \geq \frac{n}{2}$, then G is Hamiltonian. Dirac's Theorem is what is known as an **edge density condition**. The idea behind edge density conditions is that if enough edges are supplied, the more easily the next vertex can be reached to complete a Hamiltonian cycle.

Ore's Theorem: If G is a graph of order n such that $\deg x + \deg y \geq n$, for every pair of nonadjacent vertices $x, y \in V$, then G is Hamiltonian. In other words, for any pair of nonadjacent vertices we can select in the graph, the sum of their degrees must be greater than or equal to n . If this can be done for every pair of nonadjacent vertices, the graph is Hamiltonian.

Bondy and Chvátal: A graph G of order n is Hamiltonian if, and only if, $C_n(G)$ is Hamiltonian. The **k -closure** of G , denoted $C_k(G)$, is the graph obtained by recursively joining pairs of nonadjacent vertices whose degree sum is at least k , until no such pairs remain. Thus, Bondy and Chvátal's theorem states that if a graph is Hamiltonian, we can recursively join pairs of nonadjacent vertices whose degree sum is at least n , and if the resulting graph is Hamiltonian, then G is Hamiltonian.

Chvátal and Erdős: If G is a graph with connectivity k such that $\beta_0(G) \leq k$, then G is Hamiltonian. The **connectivity** of a graph G is the minimum number of vertices that would need to be removed to disconnect G . The **independence number**, $\beta_0(G)$, is the size of the largest set of independent vertices that exists in G . As Chvátal and Erdős' theorem states, if the independence number of G is less than or equal to its connectivity, then G is Hamiltonian.

2 Double Path Graphs

A graph G of order n is a **double path graph** if the edges of G can be partitioned into two sets E_1 and E_2 such that:

- (a) $E_1 \cap E_2 = \emptyset$
- (b) $E_1 \cup E_2 = E(G)$
- (c) $|E_1| = |E_2|$
- (d) $\langle E_1 \rangle$ and $\langle E_2 \rangle$ are both paths of order n .

In other words, a double path graph consists of two edge-disjoint spanning paths.

2.1 Vertex Degrees in Double Path Graphs

In a double path graph G , each of the two edge-disjoint spanning paths P_1 and P_2 has two endpoints, and two or more points that lie between the endpoints, which we will refer to as middle points. Each vertex in G has a degree of 2, 3, or 4, depending on whether it is an endpoint or a middle point on each path.

A vertex is:

- degree 2 if it is an endpoint on both paths,
- degree 3 if it is an endpoint on one path and a middle point on the other, or
- degree 4 if it is a middle point on both paths.

Thus, a double path graph must always have either:

- two 2-degree vertices,
- one 2-degree and two 3-degree vertices, or
- four 3-degree vertices,

and all other vertices must be of degree 4.

Thus, the minimum degree of a double path graph G , $\delta(G)$, is always either 2 or 3. If we recall Dirac's Theorem, we can see that the only double path graphs to meet the conditions of the theorem are of order $n \leq 6$. In fact, we will see later that all double path graphs of $n \leq 6$ are Hamiltonian. However, because $\delta(G)$ stays constant even as n increases, we cannot use Dirac's Theorem for double path graphs of $n > 6$.

2.2 Ore's Theorem in Double Path Graphs

Because every vertex in a double path graph has a degree of 2, 3, or 4, this also limits our use of Ore's Theorem. No vertex in a double path graph can ever have a degree higher than 4, so any double path graph of $n > 8$ will fail to meet the conditions of the theorem. Furthermore, the vertices with the lowest degrees in a double path graph G will always be the endpoints of the two paths P_1 and P_2 that are edge-disjoint spanning paths in G .

If G has two vertices x and y of degree 2, x and y are each endpoints of both P_1 and P_2 . There cannot be an edge in P_1 that joins x and y , and there cannot be an edge in P_2 that joins x and y , so x and y are not adjacent. If x and y are not adjacent, then G does not fit the conditions of Ore's Theorem unless the order of G is $n \leq 4$, and we will see later that there is no double path graph with two vertices of degree 2 that is of order $n \leq 4$. Therefore, Ore's Theorem cannot be used on any double path graph with two vertices of degree 2.

If G has one vertex x of degree 2 and two vertices y and z of degree 3, x is an endpoint of both P_1 and P_2 , and y and z are the other endpoints of each path. Suppose x , y , and z are all adjacent to each other, as in Figure 2.1. Let us say, without loss of generality, that P_1 has endpoints x and y , and P_2 has endpoints x and z . The edge xz must lie on P_1 , and xy must lie on P_2 . However, P_1 cannot take yz out of z , because it must pass through the rest of the vertices in the graph before it can land on y , and P_2 cannot

take yz out of y for the same reason. Thus, for G to be a double path graph, x , y , and z cannot all be adjacent.

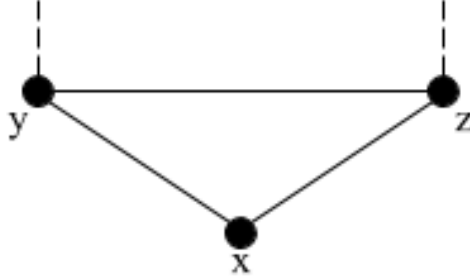


Figure 2.1: A graph with one vertex x of degree 2 and two vertices y and z of degree 3, and edges xy , xz , and yz .

If G has four vertices w , x , y , and z of degree 3, they must each be an endpoint of one of the two paths P_1 and P_2 . Because there are four vertices of degree 3, they may all be adjacent to one another only if G is K_4 , in which case they are the only vertices in the graph. Thus, if G has any vertices of degree 4, w , x , y , and z are not all adjacent.

Therefore, if G has any vertex x of degree 3 and is not K_4 , there must be at least one vertex y in G to which x is not adjacent that is of degree $\deg y \leq 3$, and thus $\deg x + \deg y \leq 6$. Since any double path graph must have either vertices of degree 3 or two vertices of degree 2, Ore's Theorem cannot be used on any double path graph of order $n > 6$.

2.3 Edge Density of Double Path Graphs

Both Dirac's and Ore's Theorems are edge density conditions. **Edge density** is the ratio of edges in a graph to the possible number of edges. In other words, the edge density of a graph G is

$$D = \frac{|E(G)|}{|E(K_n)|},$$

where K_n is the complete graph of order n . The number of edges in K_n is equal to $n(n-1)/2$. Thus, we can define edge density as

$$D = \frac{2|E(G)|}{n(n-1)}.$$

A double path on n vertices will always have exactly $2(n-1)$ edges. A single path will always have $n-1$ edges, so a double path graph must always have twice that. Thus, a double path graph will always have an edge density of

$$D = \frac{2 * 2(n-1)}{n(n-1)},$$

which we can simplify to $D = \frac{4}{n}$.

We can see that the edge density of a double path graph will grow smaller as the number of vertices increases. Edge density conditions tell us that the larger the edge density of a graph, the more likely it is to contain a Hamiltonian cycle. Thus, the larger the order of a double path graph, the harder it is to find a Hamiltonian cycle.

2.4 Double Path Graphs on Low Orders

There are no nontrivial double path graphs on $n \leq 3$ vertices.

The only double path graph on $n = 4$ vertices is K_4 :

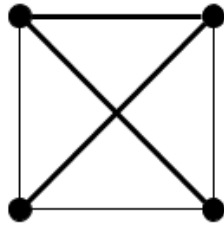


Figure 2.2: K_4 , the only double path graph of order 4.

There are only two double path graphs on $n = 5$ vertices:

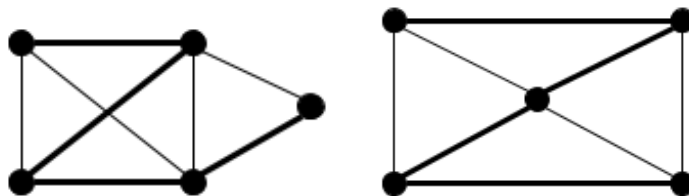


Figure 2.3: The two double path graphs of order 5.

There are only three double path graphs on $n = 6$ vertices:

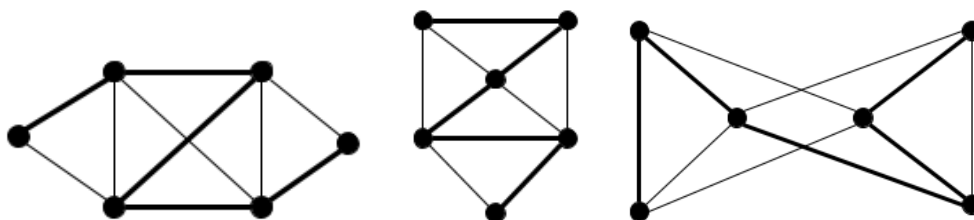


Figure 2.4: The three double path graphs of order 6.

3 Double Path Graphs and Hamiltonicity

As a double path graph necessarily contains two spanning paths, all double path graphs are by definition semi-Hamiltonian. In fact, they are semi-Hamiltonian twice over. Since a semi-Hamiltonian graph needs only one additional edge to make it Hamiltonian, it stands to reason that any double path graph is quite likely to be Hamiltonian, as it is already most of the way there.

In this section, we explore double path graphs as input to the Hamiltonian problem. We discuss under what conditions a double path graph may be Hamiltonian, and under what conditions it is not Hamiltonian.

3.1 Constructing a Hamiltonian Cycle from a Double Path

There are many ways that a spanning cycle may be constructed from the two spanning paths that make up a double path graph.

3.1.1 Using A Path and One Additional Edge

Let G be a double path graph on paths P_1 and P_2 . Let x and y be the endpoints of P_1 . If P_2 contains the edge xy , then G is Hamiltonian. The cycle $P_1 + xy$ is a spanning cycle, because P_1 is a spanning path, and x and y are its endpoints.

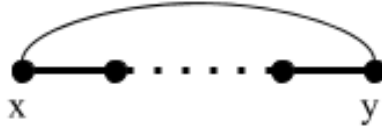


Figure 3.1: P_1 , with endpoints x and y , joined by xy , which lies on P_2 .

Of course, not every double path graph has such a convenient edge. Figure 3.2, shown below, does not fit this condition. The endpoints of P_1 are a and g , and there is no edge ag . The endpoints of P_2 are a and c , and there is no edge ac . However, Figure 3.2 is still Hamiltonian. The walk $a, b, c, d, g, h, f, e, a$ is a spanning cycle. Indeed, there are multiple spanning cycles in Figure 3.2. So we do not necessarily need the one edge e that creates a cycle $P_1 + e$ or $P_2 + e$ in order for G to be Hamiltonian.

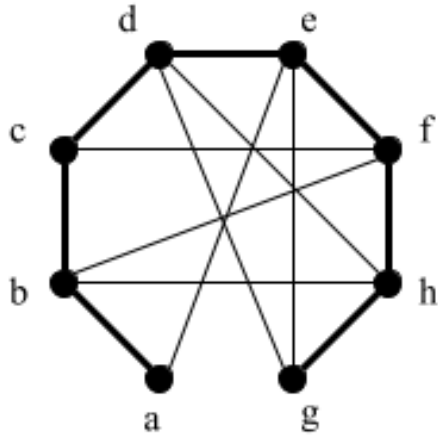


Figure 3.2: A double path graph with multiple spanning cycles.

This illustrates the primary concept of this thesis, that if we have two

spanning paths, there may be several ways to piece them together to form a spanning cycle.

3.1.2 Removing An Edge From One Path

One way to construct a spanning cycle from two spanning paths, when the convenient edge described in the previous section is not present, is to start with one path, P_1 , and remove one edge. Then, we may be able to use the edges present in P_2 to bridge the gap this creates. We can use this method to create a Hamiltonian cycle in Figure 3.2. Let us consider P_1 to be the path a, b, c, d, e, f, h, g , and P_2 to be the path a, e, g, d, h, b, f, c . If we consider only P_1 , we can see no way to close the gap left between a and g , because there is no path ag . However, if we remove the edge de , we now have two gaps in the cycle that we can close together with edges ae and dg .

We can generalize this to work for any double path graph.

3.2 Double Path Graphs without Spanning Cycles

Because there are so many ways to construct a spanning cycle from two spanning paths, it is easy to fall into the assumption that any two spanning paths can create a spanning cycle. Figure 3.3 shows a double path graph that does not contain a spanning cycle.

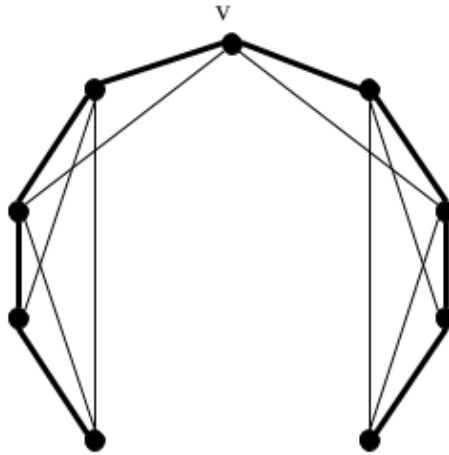


Figure 3.3: A double path graph with cut vertex v and no spanning cycles.

Interestingly, Figure 3.3 does contain a spanning circuit. However, because vertex v is a cut vertex, any circuit must pass through it twice, and thus can never be a cycle. A graph cannot have a cut vertex and also be Hamiltonian.

Because the problem in Figure 3.3 appears to be the cut vertex v , we may suppose that any double path graph that does not contain a cut vertex must be Hamiltonian. However, at least one condition exists for which this is not the case. Consider Figure 3.4.

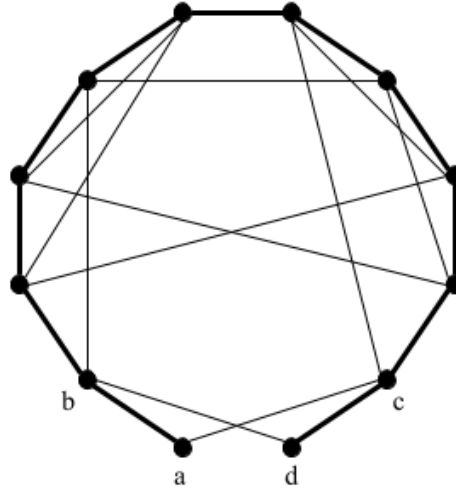


Figure 3.4: A double path graph with no cut vertices and no spanning cycles.

Figure 3.4 has no cut vertices, but does not contain a spanning cycle. We notice that Figure 3.4 does have a cut set of size 2, containing b and c , which, if removed, isolate a and d from the rest of the graph. However, a cut set of size 2 is not sufficient to render a double path graph non-Hamiltonian, as we can see from Figure 3.5.

Figure 3.4 contains two vertices of degree 2, a and d , each of which are endpoints of both P_1 and P_2 . Vertices a and d are both adjacent to the same two vertices, b and c . Because a and d have degree 2, for a spanning cycle C to exist, there must exist on C the sequence b, a, c , as well as the sequence b, d, c . Since this is impossible to achieve without passing through either b or c more than once, the spanning cycle C cannot exist on Figure 3.4.

This is true of any double path graph which contains two vertices of degree

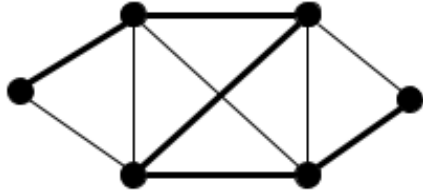


Figure 3.5: A double path graph with a cut set of order 2 and a spanning cycle.

2, wherein both vertices of degree 2 are adjacent to the same 2 vertices. However, such a configuration is only possible on double path graphs with order $n \geq 8$.